VisionPong

By: Aneesh Pabolu

A Computer Vision based approach for motion controlled video games

Demo

Systems Lab 2026 -Yilmaz

Introduction & Novelty

There have been many motion-tracking video games where a digital avatar moves based on a player's physical movement in the real world. These games work by taking information from physical controllers or sensors. Common games including Kinect Sports, 1-2 Switch, and Mario Party. However, motion-controlled games typically require expensive hardware such as game consoles. Furthermore, "vision" based motion controlled games have practically become obsolete.

In this project, I try to bring back "vision" based motion-controlled games by developing a framework for these video games to run directly off your computer and track your movement using your webcam. I take inspiration from the Xbox Kinect Sports Table Tennis game and try to replicate it using no extra hardware except a computer and a live video feed from a webcam.

Running these games locally on your computer can increase the access to motion-controlled games and improve research relating to vision based digital avatar control.

Method

As shown in *Fig. 1*, a Python script captures images from a webcam, which are then processed by MediaPipe to identify the coordinates of key body landmarks on a frame-by-frame basis.

Then, frame-by-frame, coordinates are then sent to a local port on the user's computer as a JSON text.

Using a C# script, Unity reads the values of all the important body landmark coordinates and stores them in a public class object.

Finally, as shown in *Fig. 2*, this information is then extracted by task specific C# scripts in Unity which transform those body joint positions in the digital simulation in real-time.

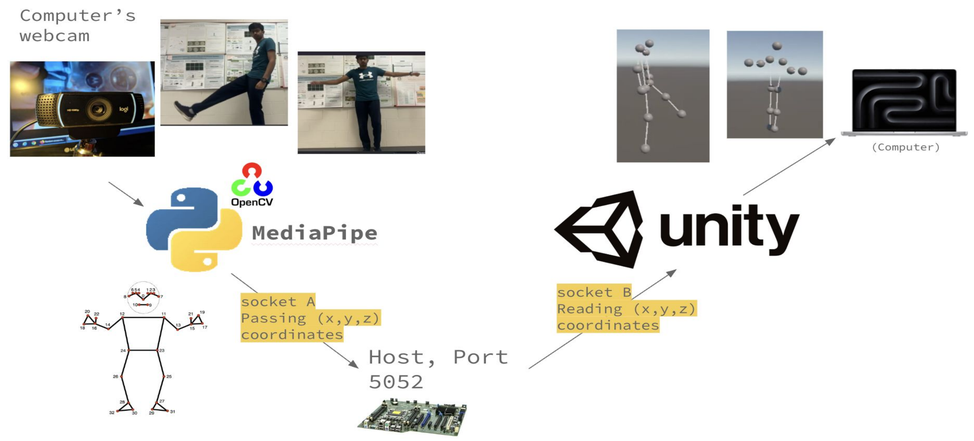


Fig. 1: Diagram highlighting the general method

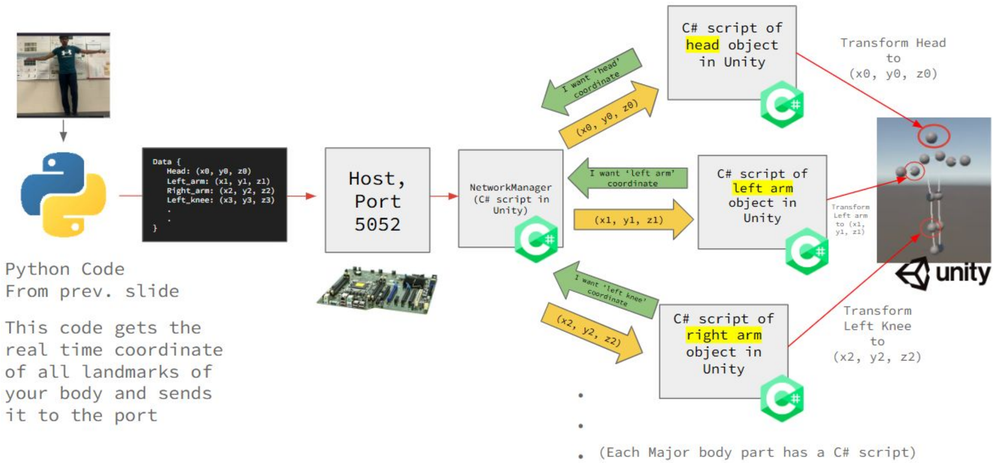


Fig. 2: Diagram highlighting how Unity receives landmark coordinates and updates the digital avatar's position

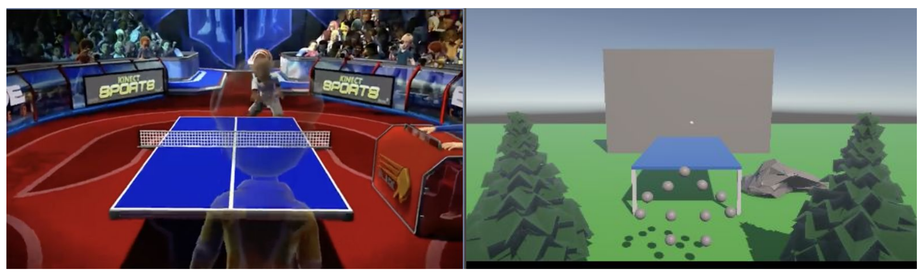


Fig. 3: Xbox Kinect Table Tennis game environment (left) shown in comparison to VisonPong's game environment (right)

Results

My project accurately gets real-time coordinates of 13 human joints with respect to the human body and replicates those proper joint positions in the Unity game scene.

Basic physics interactions between the avatar and the environment have also been implemented. This demonstrates the broader potential and extensibility of the project. A side by side comparison can be seen between Xbox Kinect's Table Tennis game environment and VisionPong's game environment in *Fig. 3.*

Impact & Future Work

VisionPong demonstrates how motion-controlled games can reach to a larger audience by eliminating the need for expensive hardware, thereby making non-sedentary indoor physical activities more accessible and appealing.

Future work includes decreasing jitter, fully implementing proper Table Tennis physics, testing various webcams and webcam positions and computers to sample game performance, as well as adding surround sound capabilities for a more immersive game experience. These enhancements will further broaden the game's appeal and accessibility.